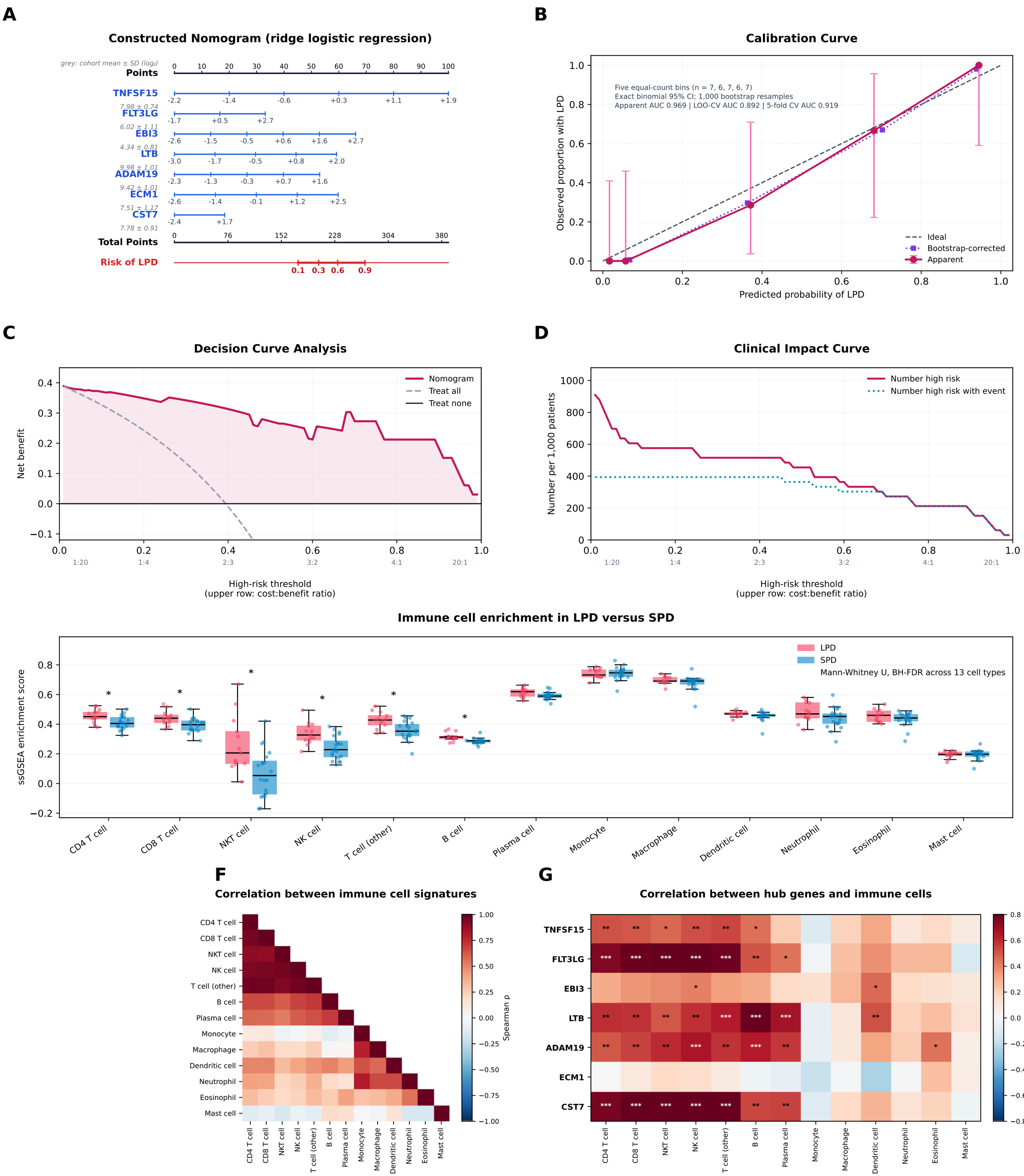


Figure 5. Nomogram based on the seven validated hub genes and its immune context. (A) Nomogram from a ridge logistic regression model fitted on GSE125498 (n = 33); gene axes are cohort z-scores and the italic grey text is the raw $\log_2$ cohort mean ± SD. (B) Apparent and bootstrap-corrected calibration in five equal-count bins with exact binomial 95% CIs, using 1,000 bootstrap resamples; apparent, LOOCV, and 5-fold CV AUCs were 0.969, 0.892, and 0.919, respectively. (C) Decision curve analysis. (D) Clinical impact curve. (E) ssGSEA scores in LPD (red) versus SPD (blue); Mann-Whitney U, BH-FDR adjusted across the 13 cell types. (F) Spearman correlation among immune signatures. (G) Spearman correlation between hub genes and immune signatures; BH-FDR adjusted across all 91 gene-cell-type tests. *FDR < 0.05, **FDR < 0.01, ***FDR < 0.001.